

End Semester (Back Paper) Examination 2024

Name of the course: B.Tech

Name of the paper: Engineering Mathematics II

Time: 3 hours

Semester: II

Paper Code: TMA - 201

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b and c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each sub part carries 10 marks.

Q.1	(10×2=20 Marks)	
(a)	Solve: $(x^2 + y^2)dx + 2xydy = 0$.	CO: 1
(b)	Solve: $(D^2 - 2D + 1)y = x \sin x$.	
(c)	Solve by the method of variation of parameters: $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$.	
Q.2	(10×2=20 Marks)	
(a)	Evaluate $L[t^2 e^{2t} \sin t]$.	CO: 2
	Using convolution theorem prove that	
(b)	$L^{-1}\left\{\frac{s^2}{(s^2 + a^2)(s^2 + b^2)}\right\} = \frac{a \sin at - b \sin bt}{a^2 - b^2}$	
(c)	Solve the differential equation using Laplace transform method. $\frac{d^2x}{dt^2} + 9x = \cos 2t, \quad x(0) = 1, \quad x\left(\frac{\pi}{2}\right) = -1$	
Q.3	(10×2=20 Marks)	
(a)	Expand the function $f(x) = x \sin x$, as a Fourier series in the interval $-\pi \leq x \leq \pi$.	CO: 3
(b)	Find the Fourier series expansion of the periodic function of period 2π . $f(x) = e^x, \quad 0 < x < 2\pi$	
(c)	Obtain the Fourier series expansion of $f(x) = \begin{cases} x, & 0 < x < \pi \\ -x, & -\pi < x < 0 \end{cases}$	
Q.4	(10×2=20 Marks)	
(a)	Solve $\frac{\partial^2 z}{\partial x^2} - 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = e^{2x-y} + e^{x+y} + \cos(x+2y)$	CO: 4
(b)	Solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$; $u(x, 0) = 6e^{-3x}$	
(c)	Find the temperature in a bar of length 2 whose ends kept at zero and lateral surface insulated if initial temperature is $\sin \frac{\pi x}{2} + 3 \sin \frac{5\pi x}{2}$.	
Q.5	(10×2=20 Marks)	
(a)	Prove that: $\int_{-1}^1 P_m(x) P_n(x) dx = 0$, if $m \neq n$	CO: 5
(b)	Prove that: $x J_n'(x) = n J_n(x) - x J_{n+1}(x)$	
(c)	Prove that: $(2n+1)x P_n = (n+1)P_{n+1} + n P_{n-1}$	